Week 6 reading guide and worked examples

Weekly overview

Required learning outcomes

By the end of this week, you should be able to:

- solve unconstrained multivariable optimisation problems and distinguish candidates, local optima and global optima using necessary and sufficient conditions (M9); and
- work with independent and identically distributed samples, sample means and sample variances, use elementary sampling distributions, and distinguish and apply the law of large numbers and the central limit theorem (S6).

Prerequisites and retrieval

For optimisation, retrieve column gradients (M6), implicit differentiation (M7) and definiteness, convexity and concavity (M8). For sampling, retrieve algebra (M2), standard distributions (S3), expectation and variance (S4), and conditional moments (S5). Questions 9–11 are a cumulative block on M6–M8 and S3–S5. Complete those three parts without consulting earlier notes.

Student materials

Use the reading guide for the exact written sections, locators and skip instructions. Then study and reproduce the original worked examples. Attempt all parts of the exercise sheet before consulting the progressive hints or complete solutions. Finish with the 25-minute self-test, mark it, and update your error log by outcome ID.

The external readings remain on their authors' sites. No external exercise is reproduced here. No video, applet, code or specialist software is required.

Suggested study sequence

1. Review column gradients, Hessian definiteness, normal means and variances.
2. Complete the exact reading selections and their closed-source summaries.
3. Cover each worked solution, reproduce the example from a blank page, and explain which conditions make the conclusion valid.
4. Attempt all 16 scorable exercise parts. Use hints only after a serious attempt and stop as soon as a hint supplies a useful next step.
5. Mark from the full solutions. Record the first cause of each error under M9, S6, or the relevant retrieval outcome.
6. Take the self-test without notes. Use the final five minutes of the weekly review block to record errors and schedule a blank-page reattempt two or three days later.

Workload and required outputs

The times below are approximate planning estimates and have not yet been verified by students.

Component	Target time	Required output
Required external reading	3 h 5 min	Two brief original summaries with the relevant conditions and conventions
Institutional notes and worked-example reproduction	1 h 10 min	Four examples reproduced from a blank page
Week 6 problem sheet	4 h 30 min	All 16 scorable parts attempted and marked

Component	Target time	Required output
Self-test, immediate marking and error-log update	30 min	Four self-test parts attempted and marked; errors labelled by outcome
Core total	**9 h 15 min**	Completed Week 6 study cycle

The problem sheet has 11 numbered questions, 16 scorable parts, 100 marks and a 270-minute target. Across marks, 32% is foundation, 58% is standard and 10% is stretch. Parts whose primary demand is explanation, derivation or synthesis carry 44% of the marks.

End-of-week check

Before moving on, make sure that you can state why a stationary condition is only necessary in its usual interior setting, identify evidence that upgrades a local conclusion to a global one, write the course definitions of $\bar{X}$ and S^2, and explain the different jobs of the law of large numbers and the central limit theorem.

Required reading route

How to use this route

Complete the two required written selections below. Read only the named sections and follow every skip instruction. Allow 185 minutes for the external reading and closed-source summaries. The separate 70-minute allocation in worked_examples.md is for institutional notes and example reproduction.

If equation rendering or assistive-mathematics access prevents you from using a linked page, skip that page and use this course's original worked examples, hints and solutions for the same outcome. Report the access problem to the course team. To report a broken, inaccessible or unavailable required source, email cristiano.cantore@uniroma1.it with the week, item or source locator. You may request this confidentially without disclosing disability or other sensitive personal information. The programme Coordinator will provide an accessible replacement PDF or an approved alternative reading. You will not be assessed on source-specific wording.

Unconstrained multivariable optimisation (`M9`)

Resource `MATH-M9-OSB-REQ`; source `OSBORNE_MM`. In Martin J. Osborne, *Mathematical Methods for Economic Theory*, read:

- Section 4.2, "Optimization: definitions";
- Section 4.3, "Existence of an optimum";
- Section 5.1 on necessary conditions;
- Section 5.2 on local optima; and
- Section 5.3 on global optima.

Skip: the proof in Section 4.3; topology beyond the operational statement that a continuous function on a non-empty closed and bounded feasible set attains a maximum and a minimum; interactive graphs; and one-variable repetition already assigned under `M3`. Approval is limited to this exact external-link route: do not copy, excerpt, adapt, translate, reformat or embed the source's text, diagrams, exercises or solutions.

Target reading time: 85 minutes.

After closing the source, write an original half-page summary that distinguishes a candidate, a strict local optimum, a local optimum and a global optimum. State separately: the interior first-order necessary condition, the definite-Hessian local sufficient conditions, a curvature condition that can support a global conclusion, and an operational existence condition.

Random samples, sampling distributions, LLN and CLT (S6)

Resource STAT-S6-REQ-PISHRO; source PISHRO_NIK. In Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*, read the following non-contiguous sections in order:

- §7.1.1, "Law of Large Numbers";
- §7.1.2, "Central Limit Theorem";
- §7.1.3, "Solved Problems", only the registered problems;
- §8.1.1, "Random Sampling", from "Random Sampling" through "Sample Mean" only; and
- §8.2.2, "Point Estimators for Mean and Variance", only the definition and elementary sample-variance discussion.

Skip: in §8.1.1 stop before "Order Statistics"; in §8.2.2 stop after the elementary sample mean and sample variance. Do not use simulations, MATLAB, other software or material on later estimation or inference. Approval is limited to this exact external-link route: do not copy, excerpt, adapt, translate, reformat or embed the source's explanations, examples, questions or solutions.

Target reading time: 100 minutes.

After closing the source, write the course conventions

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Then write an original comparison of the law of large numbers and the central limit theorem. Include the i.i.d. assumption, the population mean and finite variance conditions used in this week, the quantity that converges under the law of large numbers, and the correctly standardised quantity whose distribution is approximated by the central limit theorem.

Reading output check

Your two summaries should make every condition visible. Do not describe the law of large numbers as a claim of approximate normality, and do not describe the central limit theorem as saying that the observations themselves become normal. Keep source wording out of the summaries: the output must be your own account of the assigned ideas.

Worked examples

These original institutional notes and examples support M9 and S6. Allow about 70 minutes to study and reproduce them. Cover each **Solution**, follow the **Method**, and reproduce the work from a blank page before reading the **Interpretation**.

Optimisation conditions

Let $f : \mathbb{R}^n \to \mathbb{R}$. A point is a candidate when it has survived the relevant necessary-condition and feasibility checks; it is not thereby an optimum. If f is differentiable and $\mathbf{x}^*$ is an interior local optimum, then the first-order necessary condition is

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$

If f is twice continuously differentiable near a stationary point, a positive definite Hessian is sufficient for a strict local minimum and a negative definite Hessian is sufficient for a strict local maximum. An indefinite Hessian identifies a saddle point. A semidefinite or zero Hessian does not generally settle the classification.

Local and global conclusions are different. Strict concavity on a convex domain makes any stationary point the unique global maximiser, provided the point exists in the domain. More generally, a continuous objective on a

non-empty closed and bounded feasible set attains both a maximum and a minimum. This operational existence statement is enough here; no topological proof is required.

Example 1 — classifying stationary points (M9)

Let

$$f(x, y) = (x^2 - 1)^2 + y^2.$$

Find and classify every stationary point.

Method. Set the column gradient equal to zero. Evaluate the Hessian at each candidate and classify it by definiteness.

Solution. The column gradient and Hessian are

$$\nabla f(x, y) = \begin{bmatrix} 4x(x^2 - 1) \\ 2y \end{bmatrix}, \qquad \mathbf{H}_f(x, y) = \begin{bmatrix} 12x^2 - 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

The stationary equations give $x \in \{-1, 0, 1\}$ and $y = 0$. At $(-1, 0)$ and $(1, 0)$, the Hessian is $\operatorname{diag}(8, 2)$, which is positive definite, so both points are strict local minima. At $(0, 0)$, the Hessian is $\operatorname{diag}(-4, 2)$, which is indefinite, so the origin is a saddle point.

Interpretation. The first-order condition produces three candidates, and the Hessian distinguishes their local behaviour. The expression defining f is non-negative and equals zero only at $(-1, 0)$ and $(1, 0)$, so those two points are also global minimisers. That global conclusion comes from the objective on the whole domain, not from evaluating the Hessian only at the candidates.

Example 2 — upgrading a local result to a global one (M9)

Let

$$g(x, y) = \log x + \log y - x - y, \qquad (x, y) \in (0, \infty)^2.$$

Find the global maximiser and maximum value.

Method. Solve the stationary equations. Use the Hessian to establish strict concavity on the whole domain, then evaluate the objective.

Solution. We have

$$\nabla g(x, y) = \begin{bmatrix} 1/x - 1 \\ 1/y - 1 \end{bmatrix}, \qquad \mathbf{H}_g(x, y) = \begin{bmatrix} -1/x^2 & 0 \\ 0 & -1/y^2 \end{bmatrix}.$$

The unique stationary point is $(1, 1)$. For every point in the domain and every non-zero vector $[a, b]^\mathsf{T}$,

$$\begin{bmatrix} a & b \end{bmatrix} \mathbf{H}_g(x, y) \begin{bmatrix} a \\ b \end{bmatrix} = -\frac{a^2}{x^2} - \frac{b^2}{y^2} < 0.$$

Thus the Hessian is negative definite everywhere, so g is strictly concave on the convex domain $(0, \infty)^2$. Hence the stationary point is the unique global maximiser. The maximum value is

$$g(1, 1) = -2.$$

Interpretation. Negative definiteness at a point gives a strict local maximum. Negative definiteness everywhere on the convex domain supplies the global strict-concavity argument. The domain and the curvature calculation are different from merely classifying a constant quadratic Hessian.

Random samples and sampling distributions

Write

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} F$$

only when the observations are both independent and identically distributed. If their common mean and variance are μ and $\sigma^2 < \infty$, then

$$\mathbb{E}[\bar{X}] = \mu, \qquad \operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

The sample variance convention is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

For a normal random sample,

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

exactly, and

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

For a non-normal i.i.d. sample with finite, positive variance, the central limit theorem instead gives the large-n approximation

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \overset{\text{approximately}}{\sim} \mathcal{N}(0,1).$$

Example 3 — an exact sampling distribution for a count (S6)

Suppose

$$X_1, \dots, X_{10} \overset{\text{iid}}{\sim} \text{Uniform}(0,1),$$

and let

$$K := \sum_{i=1}^{10} \mathbf{1}\{X_i > 3/4\}.$$

Find the exact distribution of K, and calculate $\mathbb{P}(K = 0)$.

Method. Identify the distribution of each indicator, then use independence to recognise the distribution of their sum.

Solution. For each observation,

$$\mathbb{P}(X_i > 3/4) = \frac{1}{4}.$$

The indicators are independent Bernoulli(1/4) random variables. Hence

$$K \sim \text{Binomial}\left(10, \frac{1}{4}\right)$$

exactly, and

$$\mathbb{P}(K = 0) = \binom{10}{0}\left(\frac{1}{4}\right)^0\left(\frac{3}{4}\right)^{10} = \left(\frac{3}{4}\right)^{10}.$$

Interpretation. A sampling distribution need not be normal and need not belong to the sample mean. Here the statistic counts observations with a specified property, and its exact law follows from the i.i.d. sampling model.

Example 4 — independence is not identical distribution (S6)

Suppose X_1, X_2, X_3 are independent and

$$X_1 \sim \mathcal{N}(2, 1), \qquad X_2 \sim \mathcal{N}(4, 4), \qquad X_3 \sim \mathcal{N}(6, 9).$$

For $A := (X_1 + X_2 + X_3)/3$, find the exact distribution of A. Decide whether these observations form an i.i.d. random sample and whether the course's i.i.d. LLN or CLT can be invoked from the stated information.

Method. Use independence to combine normal variables, but check independence and identical distribution as separate conditions.

Solution. Independence gives

$$\mathbb{E}[A] = \frac{2 + 4 + 6}{3} = 4$$

and

$$\text{Var}(A) = \frac{1 + 4 + 9}{3^2} = \frac{14}{9}.$$

Because a linear combination of independent normal variables is normal,

$$A \sim \mathcal{N}\left(4, \frac{14}{9}\right)$$

exactly. The observations are independent, but they are not identically distributed: their means and variances differ. Therefore they cannot be written as $X_1, X_2, X_3 \overset{\text{iid}}{\sim} F$, and the course's i.i.d. LLN or CLT does not follow from these three observations and assumptions.

Interpretation. Independence allows variances to add in the calculation of A, but it does not make the marginal distributions identical. Check both words in "i.i.d." before using a random-sample formula or a limiting theorem.

Reproduction check

Before starting the exercise sheet, reproduce one complete optimisation classification and one sampling standardisation. Say aloud which assumptions support each local, global, exact or approximate conclusion.